

ENME 441

PCB Design with KiCad

schematic → layout → fabrication

KiCad

KiCad is a free, open-source Electronic Design Automation (EDA) suite for designing schematics and printed circuit boards

1

Schematic Editor (Eeschema)

Capture your circuit as symbols connected by nets.

2

Footprint & symbol libraries

Thousands of built-in parts, plus custom and vendor libraries.

3

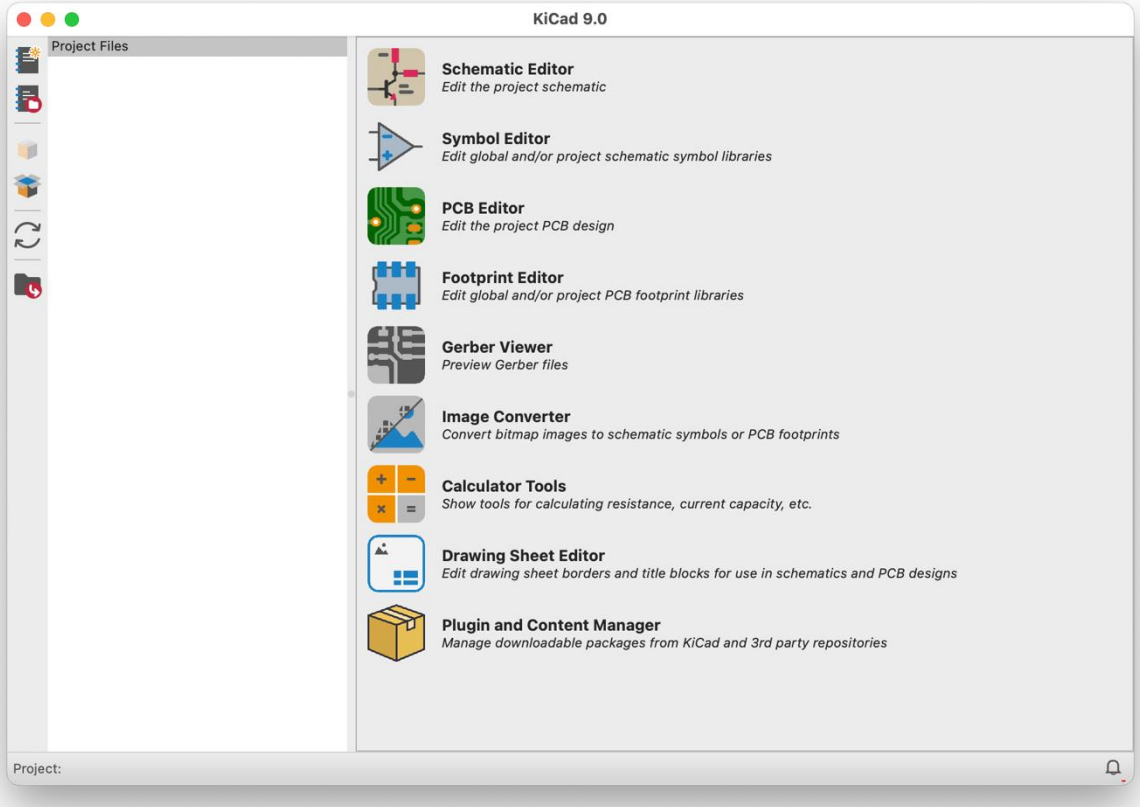
PCB Editor (Pcbnew)

Place footprints and route copper on a physical board.

4

3D viewer & fabrication export

Preview the assembled board and output manufacturing files.



The KiCad Workflow



1 · Schematic

Define all components and their connections.

2 · Layout

Position & route the components on the physical board, add ground plane(s), and perform a design rule check

3 · Export

Generate files for the PCB fab foundry.

01

Schematic Capture

Define the circuit components and electrical connections



Schematic Editor Essentials

1

Place symbols

Drop components from the libraries into the schematic.

2

Wire the nets

Connect pins with wires; place junctions where wires connect.

3

Power & ground

Use power symbols (VCC, GND) instead of hand-wiring every rail. This will also make it easier to add ground zones during PCB layout

4

Labels & annotation

Name nets with labels; annotate to give every part a unique reference (R1, U2…).

Symbols vs. Footprints



Symbol

Defines the component pins and electrical connections. Allows creation of a netlist (connections list) for use in PCB layout.

Used during **schematic design**



Footprint

Defines the physical land pattern for the component (copper pad positions and overall part dimensions).

Used during **PCB layout**

Schematic **symbols** and PCB layout **footprints** are independent entities

A specific footprint must be assigned to each symbol to allow for PCB wiring

Allows for flexible many-to-one or one-to-many relationships

Work Fast: Key Hotkeys

eeschema — hotkeys

A	**A**dd (place a symbol)
W	draw a **W**ire
P	place **P**ower symbol
L	place a net **L**abel
R	**R**otate selection
M	**M**ove (lose wires)
G	dra**G** (keep wires)
C	**C**opy
Del	delete

Hover, then press

KiCad hotkeys act on whatever is under the cursor — no need to click on the component.

Same keys in both editors

Most hotkeys behave consistently in the schematic and board editors.

Remap if desired

Preferences → Hotkeys lets you customize the full list.

Electrical Rules Check (ERC)

Run ERC before moving to layout to catch connectivity mistakes early:

Inspect → Electrical Rules Checker

Select an error in the ERC dialog to highlight the error in the schematic

Unconnected pins

A pin left floating is often a forgotten wire or missing junction.

Conflicting outputs

Two output pins tied together

Power / ground flag

PWR_FLAG is a special symbol to tell the ERC where power comes from. Add one PWR_FLAG for each power symbol (GND, +5V, VCC, 3V3, etc.) on the single net wire where each power or ground is applied to the circuit.

Unassigned references

Symbols named R? / U? require annotation before generating a netlist.

02

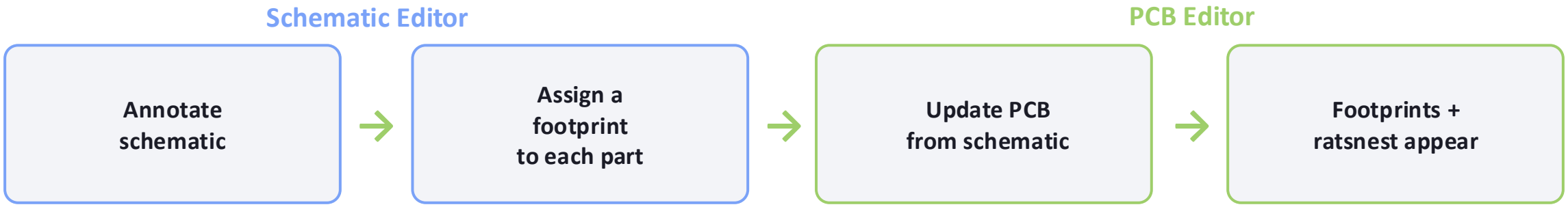
PCB Layout

Define physical positions and connections on the PCB



From Schematic to Board

KiCad pushes the finished schematic into the board editor for you.



Ratsnest — thin “rubber-band” lines showing the physical connections required for routing

KiCad PCB Layers

	Layer	Role	What goes on it	Fabricated?
	F.Cu / B.Cu	Copper	Tracks, pads, copper zones (ground and power planes) and copper text on the front and bottom layers	**Yes**
	In1.Cu, In2.Cu …	Inner copper	Extra copper layers on 4+ layer boards, often solid ground or power planes	**Yes (multilayer)**
	F.Mask / B.Mask	Solder mask	Openings in the mask where pads are exposed; generated from the pads	**Yes**
	F.Silkscreen / B.Silkscreen	Printed legend	Part outlines, reference designators, polarity marks, labels and logos	**Yes**
	Edge.Cuts	Board outline	Board outline, internal cutouts and slots the fab routes along	**Yes (shape)**
	F.Paste / B.Paste	Solder paste	Stencil openings for SMD pads, used for reflow assembly	Stencil only
	F.Adhesive / B.Adhesive	Glue	Glue dots that hold SMD parts for wave soldering; rarely used	Assembly only
	F.Courtyard / B.Courtyard	Keep-out	Space each part needs; DRC flags overlapping courtyards	No
	F.Fab / B.Fab	Assembly	Part body outlines, values and pin-1 marks for assembly drawings	No
	User layers, Margin	Notes	Drawings, comments, Eco layers and the placement margin; documentation only	No

Vias and through-hole pads span every copper layer; their holes live in the drill file, not a Gerber layer. **A fab needs the copper, mask, silkscreen and Edge.Cuts Gerbers plus the drill file; paste is only for a stencil.**

Board Setup: Layers & Rules

1

Choose a layer count

Two copper layers (top + bottom) is typical for simple boards (more for dense designs)

2

Set the board outline

Draw the physical shape on the Edge.Cuts layer

3

Define design rules

Track width, clearance, and via sizes — match to fab house requirements.

4

Key layers

F / B	Front / Back sides (2-layer PCB)
Cu	copper (traces, planes)
mask	solder mask, defines regions where copper is exposed (automatic on device pads)
silkscreen	ink overlay (printed over solder mask)
edge.cuts	through-board cuts (overall board boundary, holes)

Placement & Routing

- **Place first, route second**

  Arrange footprints logically — connectors at edges, decoupling caps near their ICs.

- **Follow the ratsnest**

  Route the rubber-band connections into real copper tracks.

- **Manage layers with vias**

  Drop a via to jump a track to another copper layer and avoid crossings.

- **Mind track width**

  Wider tracks can carry more current and drop less voltage — size power nets accordingly.

- **Keep return paths short**

  Route signals over continuous ground for clean, low-noise returns.

- **Keep signals away from WiFi antenna**

  Avoid potential interference by adding a robust ground plane between signal traces and the location of the ESP32 WiFi antenna

Add ground zones

- Fill unused copper with a GND zone for shielding and easy return paths.
- For a 2-layer board create ground planes on both sides.
- Add vias to connect isolated islands.
- Zones don't auto-update — refill (B) before re-running DRC
- Route signals over continuous ground for clean, low-noise returns.

Design Rules Check

Pour ground planes, then let DRC verify the board is manufacturable.

Unrouted nets

Ratsnest lines mean connections aren't finished.

Clearance violations

Tracks or pads too close together — DRC flags anything below your rule.

03

Exporting for PCB fab



Fabrication Outputs

Required:

Gerber files

One file per layer. Gerber is the universal PCB manufacturing format.

Drill files

Hole positions and sizes for vias and through-holes.

Optional:

Bill of Materials

List of all components and quantities for assembly & ordering.

Placement (CPL)

Component positions for pick-and-place assembly.

3D model / STEP

Verify fit and export for mechanical CAD.

Order via fab house

Zip all Gerber files and upload to JLCPCB, OSH Park, etc.



Lab 9 – PCB Layout

(Lab 7 due to change in lecture sequence)